

Naive Bayes

1

S1) from table

No of students = 10

Maths High = 5

Low = 5

Attendance

good = 5

Poor = 5

Internal

High = 2

Medium = 5

Poor = 3

Pass = 6

Fail = 4

S2) goal $\Rightarrow$ Predict for new student
Maths = High Attendance = Poor
Internal = Medium

S3) How to solve find class probabilities

$$P(\text{Pass}) = 6/10 = 0.6$$

$$P(\text{Fail}) = 4/10 = 0.4$$

S4) find conditional probabilities
for pass

$$P(\text{Maths} = \text{High} | \text{Pass}), P(\text{Attendance} = \text{Poor} | \text{Pass})$$

$$P(\text{Internal} = \text{Medium} | \text{Pass})$$

55) Count Values

for Pass

$$\text{Maths} = \text{High} = 5$$

$$\text{Attendance} = \text{Poor} = 2$$

$$\text{Internal} = \text{Medium} = 3$$

for fail

$$\text{Maths} = \text{High} = 0$$

$$\text{Attendance} = \text{Poor} = 3$$

$$\text{Internal} = \text{Medium} = 1$$

56) Apply Laplace Smoothing

$$\text{Maths} = 2 \text{ (High, Low)}$$

$$\text{Attendance} = 2 \text{ (good, Poor)}$$

$$\text{Internal} = 3 \text{ (High, Med, Poor)}$$

57) for Pass

$$P(\text{Maths} = \text{High} | \text{Pass}) = (5+1) / (6+2) = 6/8 = 0.75$$

$$P(\text{Attendance} = \text{Poor} | \text{Pass}) = (2+1) / (6+2) = 3/8 = 0.375$$

$$P(\text{Internal} = \text{Medium} | \text{Pass}) = (3+1) / (6+3) = 4/9 = 0.44$$

for fail

$$P(\text{Maths} = \text{High} | \text{Fail}) = (0+1) / (4+2) = 1/6 = 0.17$$

$$P(\text{Attendance} = \text{Poor} | \text{Fail}) = (3+1) / (4+2) = 4/6 = 0.67$$

$$P(\text{Internal} = \text{Medium} | \text{Fail}) = (1+1) / (4+3) = 2/7 = 0.29$$

58) Apply Naïve Bayes formula

for Pass

$$P(\text{Pass} | \text{Data}) = 0.68 \times 0.75 \times 0.375 \times 0.44$$

$$\text{Result} = 0.074$$

$$\text{for fail } P(\text{Fail} | \text{Data}) = 0.17 \times 0.4 \times 0.67 \times 0.29$$

$$\approx 0.13$$

sq) final Comparison

3

$$\text{Pass} = 0.074$$

$$\text{Fail} = 0.013$$

sq) final prediction

$$\text{Prediction} = \underline{\text{Pass}}$$

Student	Maths	Attendance	Internal	Result
S1	H	G	H	P
S2	H	G	H	P
S3	H	P	M	P
S4	H	G	M	P
S5	L	P	L	F
S6	L	P	L	F
S7	H	G	M	P
S8	L	G	L	F
S9	H	P	M	P
S10	L	P	M	F

i) Predict for Pass for Pass Maths = Low Attendance = Medium Internal = High for fail

$$\begin{aligned} \text{Maths} &= 1 = \text{Low} \\ \text{Attendance} &= P = 2 \\ \text{Internal} &= H = 2 \end{aligned}$$

$$\begin{aligned} \text{Maths} &= 4 \\ \text{Attendance} &= 3 \\ \text{Internal} &= 0 \end{aligned}$$

$$P(\text{Maths} = \text{Low} | \text{Pass}) = (1+1) / (6+2) = 2/8 = 0.25$$

$$P(\text{Attendance} = \text{Poor} | \text{Pass}) = (2+1) / (6+2) = 3/8 = 0.375$$

$$P(\text{Internal} = \text{Medium} | \text{Pass}) = (2+1) / (6+2) = 3/9 = 0.333$$

for fail

$$P(\text{Maths} = \text{High} | \text{fail}) = (4+1) / (4+2) = 5/6 = 0.833$$

$$P(\text{Attendance} = \text{Poor} | \text{fail}) = (3+1) / (4+2) = 4/6 = 0.667$$

$$P(\text{Internal} = \text{High} | \text{fail}) = (0+1) / (4+3) = 1/7 = 0.143$$

Naive Bayes

$$\begin{aligned} \text{for pass} &= 0.6 \times 0.25 \times 0.375 \times 0.333 \\ &= 0.019 \end{aligned}$$

$$\begin{aligned} \text{for fail} &= 0.4 \times 0.833 \times 0.667 \times 0.143 \\ &= 0.032 \end{aligned}$$

Prediction

fail

11)

Predict for

Maths = High

Attendance = Good

Internal = Low

for Pass

Maths = High = 5

Attendance = good = 4

Internal = Low = 0

for fail

Maths = 0

Attendance = 1

Internal = low = 3

for Pass & fail

$$P(\text{Maths} = \text{High} | \text{Pass}) = (5+1) / (6+2) = 6/8 = 0.75$$

$$P(\text{Attendance} = \text{good} | \text{Pass}) = (4+1) / (6+2) = 5/8 = 0.625$$

$$P(\text{Attendance} = \text{good} | \text{fail}) = (1+1) / (4+2) = 2/6 = 0.333$$

$$P(\text{Maths} = \text{High} | \text{fail}) = (0+1) / (4+2) = 1/6 = 0.17$$

$$P(\text{Internal} = \text{Low} | \text{pass}) = (0+1) / (6+3) = 1/9 = 0.111$$

$$P(\text{Internal} = \text{low} | \text{fail}) = (3+1) / (4+3) = 4/7 = 0.571$$

$$\text{Pass} = 0.6 \times 0.75 \times 0.625 \times 0.111$$

$$= 0.031$$

5

$$\text{fail} = 0.4 \times 0.333 \times 0.17 \times 0.571$$

$$= 0.013$$

Prediction

Pass

Predict for Maths = High, Attendance = good,
Internal = High

Pass

$$P(\text{Maths} = \text{High}) = 5$$

$$\text{Attendance} = \text{good} = 4$$

$$\text{Internal} = \text{High} = 2$$

$$P(\text{Maths} = \text{High} | \text{Pass})$$

$$= (5+1) / (6+2) = 6/8$$

$$= 0.75$$

$$P(\text{Attendance} = \text{good} | \text{Pass})$$

$$= (4+1) / (6+2) = 5/8$$

$$= 0.625$$

$$P(\text{Internal} = \text{High} | \text{Pass})$$

$$= (2+1) / (6+3)$$

$$= 3/9 = \cancel{0.333} 0.333$$

$$\text{Pass} = 0.6 \times 0.75 \times 0.625 \times 0.333 = 0.094$$

$$\text{Fail} = 0.4 \times 0.17 \times 0.29 \times 0.143 = 0.003$$

Prediction = Pass

$$\text{Maths} = 0 \text{ fail}$$

$$\text{Attendance} = 1$$

$$\text{Internal} = 0$$

$$P(\text{Maths} = \text{High} | \text{Fail})$$

$$= (0+1) / (4+2) = 1/6$$

$$= 0.17$$

$$P(\text{Attendance} = \text{good} | \text{fail})$$

$$= (1+1) / (4+2) = 2/6$$

$$= 0.29$$

$$P(\text{Internal} = \text{High} | \text{fail})$$

$$= (0+1) / (4+3)$$

$$= 1/7 = 0.143$$

Predict Math = High Attendance-Poor
Internal = Low

for Pass

Maths = 5

Attendance = 2

Internal = 0

$$\begin{aligned}P(\text{Maths} = \text{High} | \text{Pass}) \\&= (5+1) / (6+2) = 6/8 \\&= 0.75\end{aligned}$$

$$\begin{aligned}P(\text{Attendance} = \text{Poor} | \text{Pass}) \\&= (2+1) / (6+2) = 3/8 \\&= 0.375\end{aligned}$$

$$\begin{aligned}P(\text{Internal} = \text{Low} | \text{Pass}) \\&= (0+1) / (6+3) = 1/9 \\&= 0.111\end{aligned}$$

$$\begin{aligned}\text{Pass} &= 0.6 \times 0.75 \times 0.375 \times 0.111 \\&= 0.019\end{aligned}$$

$$\begin{aligned}\text{Fail} &= 0.4 \times 0.17 \times 0.67 \times 0.571 \\&= 0.026\end{aligned}$$

Prediction

Fail

for fail

Maths = 0

Attendance = 3

Internal = 3

$$\begin{aligned}P(\text{Maths} = \text{High} | \text{fail}) \\&= (0+1) / (4+2) = 1/6 \\&= 0.17\end{aligned}$$

$$\begin{aligned}P(\text{Attendance} = \text{Poor} | \text{fail}) \\&= (3+1) / (4+2) = 4/6 \\&= 0.67\end{aligned}$$

$$\begin{aligned}P(\text{Internal} = \text{Low} | \text{fail}) \\&= (3+1) / (4+3) = 4/7 \\&= 0.571\end{aligned}$$

Predict Math = Low

Attendance = Poor

7

Internal = Low

for pass

Math = 1

Attendance = 2

Internal = 0

$P(\text{Math} = \text{Low} | \text{Pass})$

$$= (1+1)/(6+2) = 2/8 = 0.25$$

$P(\text{Attendance} = \text{Poor} | \text{Pass})$

$$= (2+1)/(6+2) = 3/8 = 0.375$$

$P(\text{Internal} = \text{Low} | \text{Pass})$

$$= (0+1)/(6+3) = 1/9 = 0.111$$

$$\begin{aligned} \text{for pass} &= 0.6 \times 0.25 \times 0.375 \times 0.111 \\ &= 0.006 \end{aligned}$$

$$\begin{aligned} \text{for fail} &= 0.4 \times 0.8333 \times 0.667 \times 0.571 \\ &= 0.127 \end{aligned}$$

Prediction

Pass

for fail

Math = 4

Attendance = 3

Internal = 3

$P(\text{Math} = \text{Low} | \text{fail})$

$$= (4+1)/(4+2) = 5/6 = 0.833$$

$P(\text{Attendance} = \text{Poor} | \text{fail})$

$$= (3+1)/(4+2) = 4/6 = 0.667$$

$P(\text{Internal} = \text{Low} | \text{fail})$

$$= (3+1)/(4+3) = 4/7 = 0.571$$